

UNIT 4- HIGHER ORDER LINEAR DIFFERENTIAL EQUATIONS

Syllabus:

Linear equations of second and higher order with constant coefficients – Euler's and Legendre's linear equations – Method of variation of parameters – First order Simultaneous linear equations with constant coefficients – Applications: Electric Circuits.

Introduction

Differential equations are of fundamental importance in engineering mathematics because many physical laws and relations appear mathematically in the form of such equations. In the field of computers the differential equations are applied very well.

In applied mathematics, every geometrical or physical problem when translated into mathematical symbols give rise to a differential equation. Differential equations arise from many problems in oscillations of mechanical and electrical systems, bending of beams, etc., and as such play a very important role in all modern scientific and engineering studies.

An ordinary differential equation is that in which all the differential co-efficients have reference to a single independent variable.

Examples:

1. $\sin x \, dx + \cos y \, dy = 0$

2. $\frac{d^2x}{dt^2} + n^2x = 0$

3. $y = x \frac{dy}{dx} + \frac{x}{dy/dx} = 0$

The approach of an engineering student to the study of differential equations has got to be practical. Thus for an applied mathematician, the study of a differential equation consists of three phases:

- i) Formulation of differential equation from the given physical situation, called modelling.
- ii) Solution of this differential equation evaluating the arbitrary constants from the given conditions and
- iii) Physical interpretation of the solution.

Solution of a differential equation

A Solution of a differential equation is a relation between the variables which satisfies the given differential equation

Example:

$x = A \cos(nt + \alpha)$ is the solution of $\frac{d^2x}{dt^2} + n^2x = 0$

The **general solution** of a differential equation is that in which the number of arbitrary constants is equal to the order of the differential equation.

The **complementary function** is the part of the general solution of a linear differential equation which is the general solution of the associated homogeneous equation obtained by substituting zero for the terms not containing the dependent variable

The **particular solution** is that which can be obtained from the general solution by giving particular values to the arbitrary constants.

Linear differential equations with constant co-efficients

Linear differential equation are those in which the dependent variable and its derivatives occur in the first degree and are not multiplied together. Thus the linear differential equation of n^{th} order with constant co-efficients is of the form

$$\frac{d^n y}{dx^n} + k_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + k_n y = X \dots \dots (1)$$

Where $k_1, k_2, \dots, k_n$ are constants and X is a function of x .

Denote $D = \frac{d}{dx}, D^2 = \frac{d^2}{dx^2}$

The (1) can be written in the symbolic form

$$(D^n + k_1 D^{n-1} + \dots + k_n) y = X$$

$$f(D)y = X$$

It is the solution of the form

$y = \text{Complementary function} + \text{Particular Integral}$

$y = CF + PI$

where

CF is the complete solution of $f(D)y = 0$

And PI is the particular solution of

$$f(D)y = X$$

Rules for finding CF

Consider the equations

$$(D^n + k_1 D^{n-1} + \dots + k_n)y = 0$$

The corresponding auxillary equation is obtained by replacing D by 'm'.

$$(m^n + k_1 m^{n-1} + \dots + k_n)y = 0$$

That is, $m_1, m_2, \dots, m_n$ be the roots of this equation.

S.NO	ROOTS OF THE AUXILIARY EQUATION	COMPLEMENTARY FUNCTION
1.	All the roots are real and distinct	$C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$
2.	Real and equal roots If $m_1 = m_2 \neq m_3 \neq m_4$ If $m_1 = m_2 = m_3 \neq m_4$	$(C_1 x + C_2) e^{m_1 x} + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}$ $(C_1 x^2 + C_2 x + C_3) e^{m_1 x} + C_4 e^{m_4 x} + \dots + C_n e^{m_n x}$
3.	Complex roots If one pair of roots is imaginary and other roots are distinct i.e., $m_1 = \alpha + i\beta$ $m_2 = \alpha - i\beta, m_3 \neq m_4 \dots \neq m_n$	$e^{\alpha x} (C_1 \cos \beta x + C_2 \sin \beta x) + C_3 e^{m_3 x} + \dots + C_n e^{m_n x}$
4.	If two pairs of roots are imaginary and equal $m_1 = m_2 = \alpha + i\beta$ and $m_3 = m_4 = \alpha - i\beta$	$e^{\alpha x} ((C_1 x + C_2) \cos \beta x + (C_3 x + C_4) \sin \beta x + \dots + C_n e^{m_n x})$

Consider the equation

$$(D^n + k_1 D^{n-1} + \dots + k_n)y = 0$$

$$f(D)y = X$$

Then

$$PI = \frac{1}{f(D)} X$$

PROBLEMS

1.Solve $\frac{d^2 y}{dx^2} + 3 \frac{dy}{dx} - 40y = 0$

Solution

The auxillary equation is

$$m^2 + 3m - 40 = 0$$

$$(m + 8)(m - 5) = 0$$

$$m = -8, 5$$

$$CF = Ae^{-8x} + Be^{5x}, PI = 0$$

The solution is $y = CF = Ae^{-8x} + Be^{5x}$

2.Solve $\frac{d^2x}{dt^2} + n^2x = 0$

Solution

The auxillary equation is

$$m^2 + n^2 = 0$$

$$m = ni, -ni$$

$$\alpha = 0, \beta = n$$

$$CF = A \cos nt + B \sin nt$$

$$x = A \cos nt + B \sin nt$$

3. Solve $2 \frac{d^3y}{dx^3} - 5 \frac{d^2y}{dx^2} - 4 \frac{dy}{dx} + 3y = 0$

Solution

The auxillary solution is

$$2m^3 - 5m^2 - 4m + 3 = 0$$

$$(m + 1)(m - 3)\left(m + \frac{1}{2}\right) = 0$$

$$m = -1, 3, \frac{1}{2}$$

$$CF = Ae^{-x} + Be^{3x} + Ce^{\frac{x}{2}}$$

$$PI = 0$$

$$y = CF + PI = Ae^{-x} + Be^{3x} + Ce^{\frac{x}{2}}$$

4.Solve $\frac{d^3y}{dx^3} + 3 \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + 2y = 0$

Solution

The auxillary solution is

$$m^3 + 3m^2 + 3m + 2 = 0$$

$$(m + 2)(m^2 + m + 1) = 0$$

$$m = -2, m^2 + m + 1 = 0$$

$$m = \frac{-1 \pm \sqrt{1-4}}{2} = \frac{-1 \pm i\sqrt{3}}{2}$$

Roots are

$$m = -2, m = \frac{-1 \pm i\sqrt{3}}{2}$$

$$CF = Ae^{-2x} + Be^{-\frac{1}{2}x} \left(C \cos \frac{\sqrt{3}}{2}x + D \sin \frac{\sqrt{3}}{2}x \right)$$

$$PI = 0$$

$$y = CF + PI = Ae^{-2x} + Be^{-\frac{1}{2}x} \left(C \cos \frac{\sqrt{3}}{2}x + D \sin \frac{\sqrt{3}}{2}x \right)$$

5. Solve $\frac{d^3y}{dx^3} - \frac{d^2y}{dx^2} - \frac{dy}{dx} + y = 0$

Solution

The auxillary equation is

$$m^3 - m^2 - m + 1 = 0$$

$$(m - 1)(m^2 - 1) = 0$$

$$(m - 1)(m - 1)(m + 1) = 0$$

$$m = 1, 1, -1$$

$$y = e^x(A + Bx) + Ce^{-x}$$

6. Solve $\frac{d^2y}{dx^2} - \frac{dy}{dx} - 2y = 0, y(0) = 3, y'(0) = 0$

Solution

The auxillary equation is

$$m^2 - m - 2 = 0$$

$$(m - 2)(m + 1) = 0$$

$$m = 2, -1$$

$$CF = Ae^{2x} + Be^{-x}, PI = 0$$

$$y = Ae^{2x} + Be^{-x}$$

$$y' = 2Ae^{2x} - Be^{-x}$$

$$y(0) = 3, 3 = A + B$$

$$y'(0) = 0, 0 = 2A - B$$

Adding the two equations

$$3A = 3 \Rightarrow A = 1$$

Subtracting the two equations

$$A - 2B = -3 \Rightarrow A + 3 = 2B, B = 2$$

$$y = e^{2x} + 2e^{-x}$$

RULES AND PROBLEMS FOR FINDING THE PARTICULAR INTEGRAL

RULE 1:

When $X = e^{ax}, f(a) \neq 0$

$$PI = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}$$

PROBLEMS

1.Solve $(D^2 - 4D + 13)y = e^{2x}$

Solution

$$m^2 - 4m + 13 = 0$$

$$m = \frac{4 \pm 6i}{2} = 2 \pm 3i$$

$$CF = e^{2x}(A \cos 3x + B \sin 3x)$$

$$= \frac{1}{(D^2 - 4D + 13)} e^{2x} = \frac{e^{2x}}{9}$$

$$y = e^{2x}(A \cos 3x + B \sin 3x) + \frac{e^{2x}}{9}$$

2.Find k if the particular integral in the solution of $\frac{d^2y}{dx^2} - 4\frac{dy}{dx} - 2y = ke^{5x}$ is e^{5x}

Solution

$$\frac{d^2y}{dx^2} - 4\frac{dy}{dx} - 2y = ke^{5x}$$

$$PI = e^{5x}$$

$$e^{5x} = \frac{1}{D^2 - 4D - 2} k e^{5x}$$

$$1 = \frac{1}{3}k \Rightarrow k = 3$$

RULE 2:

When $X = e^{ax}, f(a) = 0$

$$PI = \frac{1}{f(D)} e^{ax}$$

$$= x \frac{1}{f'(a)} e^{ax}, f'(a) \neq 0$$

And if $f'(a) = 0$

$$PI = x^2 \frac{1}{f''(a)} e^{ax}, f''(a) \neq 0$$

and so on.

In other words

$$\frac{1}{D - a} e^{ax} = \frac{x e^{ax}}{1!},$$

$$\frac{1}{(D - a)^2} e^{ax} = \frac{x^2 e^{ax}}{2!}$$

$$\frac{1}{(D - a)^n} e^{ax} = \frac{x^n e^{ax}}{n!}$$

PROBLEMS

1.Solve $(D - 2)^2 y = e^{2x}$

Solution

The auxillary solution is

$$(m - 2)^2 = 0$$

$$m = 2, 2$$

$$CF = y = e^{2x}(Ax + B)$$

$$PI = \frac{1}{(D - 2)^2} e^{2x}$$

$$= x \cdot \frac{1}{2(D-2)} e^{2x} = \frac{x^2 e^{2x}}{2}$$

$$y = e^{2x}(Ax + B) + \frac{x^2 e^{2x}}{2}$$

RULE 3:

When $X = \sin(ax + b)$ or $\cos(ax + b)$

$$\text{i.e., } PI = \frac{1}{f(D^2)} \sin(ax + b)$$

Replace only D^2 by $-(a^2)$

$$PI = \begin{cases} \frac{1}{f(-a^2)} \sin(ax + b), f(-a^2) \neq 0 \\ x \frac{1}{f'(-a^2)} \sin(ax + b), f(-a^2) = 0 \end{cases}$$

The same procedure holds good for $\cos(ax + b)$

PROBLEMS:

$$1. (D^2 - 4D + 4)y = e^{2x} + \cos 2x$$

Solution

$$m^2 - 4m + 4 = 0$$

$$(m - 2)^2 = 0$$

$$m = 2(\text{twice})$$

$$C.F = y = (Ax + B)e^{2x}$$

$$PI = \frac{1}{D^2 - 4D + 4} (e^{2x} + \cos 2x)$$

$$PI_1 = \frac{1}{D^2 - 4D + 4} e^{2x}$$

$$= \frac{1}{0} e^{2x}$$

$$= x \frac{1}{2D - 4} e^{2x}$$

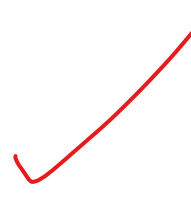
$$= x \frac{1}{0} e^{2x}$$

$$= x^2 \frac{1}{2} e^{2x}$$

$$\begin{aligned}
 PI_2 &= \frac{1}{D^2 - 4D + 4} \cos 2x \\
 &= \frac{1}{-4 - 4D + 4} \cos 2x \\
 &= \frac{1}{-4} \frac{\sin 2x}{2} = \frac{-\sin 2x}{8} \\
 y &= (Ax + B)e^{2x} + x^2 \frac{1}{2} e^{2x} \frac{-\sin 2x}{8}
 \end{aligned}$$

2.Solve $(D^2 + 6D + 8)y = e^{-2x} + \cos^2 x$

Solution

$$\begin{aligned}
 m^2 + 6m + 8 &= 0 \\
 (m + 4)(m + 2) &= 0 \\
 m &= -4, -2 \\
 C.F = y &= Ae^{-4x} + Be^{-2x} \\
 PI &= \frac{1}{D^2 + 6D + 8} (e^{-2x} + \cos^2 x) \\
 PI_1 &= \frac{1}{D^2 + 6D + 8} e^{-2x} \\
 &= \frac{1}{(-2)^2 + 6(-2) + 8} e^{-2x} \\
 &= x \frac{1}{2D + 6} e^{-2x} = x \frac{1}{2} e^{-2x} \\
 PI_2 &= \frac{1}{D^2 + 6D + 8} \cos^2 x \\
 &= \frac{1}{D^2 + 6D + 8} \left(\frac{1 + \cos 2x}{2} \right) \\
 &= \frac{1}{2} \frac{1}{D^2 + 6D + 8} e^{0x} + \frac{1}{2} \frac{1}{D^2 + 6D + 8} \cos 2x \\
 &= \frac{1}{2} \frac{1}{8} e^{0x} + \frac{1}{4} \frac{1}{3D + 2} \cos 2x \\
 &= \frac{1}{16} + \frac{1}{4} \frac{3D - 2}{9D^2 - 4} \cos 2x \\
 &= \frac{1}{16} + \frac{1}{4} \frac{-6 \sin 2x - 2 \cos 2x}{-40}
 \end{aligned}$$


$$= \frac{1}{16} + \frac{3 \sin 2x + \cos 2x}{80}$$

$$y = Ae^{-4x} + Be^{-2x} + x \frac{1}{2} e^{-2x} + \frac{1}{16} + \frac{3 \sin 2x + \cos 2x}{80}$$

RULE 4:

When $X = x^n$, where n is a positive integer,

$$PI = \frac{1}{f(D)} x^n$$

$$= [f(D)]^{-1} x^n$$

Expand $f(D)$ in ascending powers of D by using the binomial series after rewriting $f(D) = [1 \pm g(D)]$

Since $(n+1)$ th and higher order derivatives of are zero, we need not consider terms beyond D^n in the expansion.

The final binomial series and derivatives will be useful.

$$(1 - x)^{-1} = 1 + x + x^2 + \dots$$

$$(1 + x)^{-1} = 1 - x + x^2 - \dots$$

$$(1 - x)^{-2} = 1 + 2x + 3x^2 + \dots$$

$$(1 + x)^{-2} = 1 - 2x + 3x^2 - \dots$$

$$D(x^3) = 3x^2$$

$$D^2(x^3) = 3 \cdot 2 \cdot x$$

$$D^3(x^3) = 3 \cdot 2 \cdot 1$$

$$D^n(x^3) = n!$$

PROBLEMS

1. $(D^2 + 5D + 4)y = x^2 + 7x + 9$

SOLUTION

The auxiliary equation is

$$m^2 + 5m + 4 = 0$$

$$(m + 4)(m + 1) = 0$$

$$m = -4, -1$$

$$y = C.F = Ae^{-4x} + Be^{-x}$$

$$\begin{aligned}
PI &= \frac{1}{D^2 + 5D + 4} (x^2 + 7x + 9) \\
&= \frac{1}{4(1 + \frac{(D^2 + 5D)}{4})} (x^2 + 7x + 9) \\
&= \frac{1}{4} [1 + \frac{(D^2 + 5D)}{4}]^{-1} (x^2 + 7x + 9) \\
&= \frac{1}{4} [1 - \frac{(D^2 + 5D)}{4} + \frac{(D^2 + 5D)^2}{4} - \dots] (x^2 + 7x + 9) \\
&= \frac{1}{4} [1 - \frac{D^2}{4} - \frac{5D}{4} + \frac{D^4}{16} + \frac{25D^2}{16} - \dots] (x^2 + 7x + 9) \\
&= \frac{1}{4} [x^2 + 7x + 9 - \frac{1}{2} - \frac{10x}{4} - \frac{35}{4} + \frac{50}{16}] \\
&= \frac{1}{32} [8x^2 + 36x + 23]
\end{aligned}$$

2. $D^2(D^2 + 4)y = 96x^2$

SOLUTION

The axillary equation is

$$m^2(m^2 + 4) = 0$$

$$m^2 = 0 \Rightarrow m = 0$$

$$m^2 + 4 = 0$$

$$m^2 = -4$$

$$m = \pm 2i$$

$$m = 0, 0, \pm 2i$$

$$y = CF = (Ax + B)e^{0x} + e^{0x}[C \cos 2x + D \sin 2x]$$

$$\begin{aligned}
PI &= \frac{1}{4D^2(1 + \frac{D^2}{4})} 96x^2 \\
&= 96 \frac{1}{4D^2} (1 + \frac{D^2}{4})^{-1} x^2 \\
&= \frac{24}{D^2} \left(1 - \frac{D^2}{4} + \frac{D^4}{16} - \dots \right) x^2 \\
&= \frac{24}{D^2} \left(x^2 - \frac{D^2(x^2)}{4} \right)
\end{aligned}$$

$$\begin{aligned}
&= \frac{24}{D^2} \left(x^2 - \frac{1}{2} \right) \\
&= 24 \left[\frac{1x^2}{D^2} - \frac{1}{2D^2} \right] \\
&= 2x^4 - 6x^2
\end{aligned}$$

$$y = (Ax + B)e^{0x} + e^{0x}[C \cos 2x + D \sin 2x] + 2x^4 - 6x^2$$

3.Solve $(D^3 - D^2 - 6D)y = x^2 + 1$

Solution

The auxillary equation is

$$\begin{aligned}
m^3 - m^2 - 6m &= 0 \\
m(m^2 - m - 6) &= 0 \\
m(m - 3)(m + 2) &= 0 \\
m = 0, m = 3, m = -2 \\
y = CF &= (Ae^{0x}) + (B)e^{-2x} + Ce^{3x} \\
PI &= \frac{1}{D^3 - D^2 - 6D} (x^2 + 1) \\
&= \frac{1}{-6D - D^2 + D^3} (x^2 + 1) \\
&= -\frac{1}{6D + D^2 - D^3} (x^2 + 1) \\
&= -\frac{1}{6D \left[1 - \frac{D^3 - D^2}{6D} \right]} (x^2 + 1) \\
&= -\frac{1}{6D} \left[1 - \frac{D^3 - D^2}{6D} \right]^{-1} (x^2 + 1) \\
&= -\frac{1}{6D} \left[1 + \frac{D^2 - D}{6} + \left(\frac{D^2 - D}{6} \right)^2 + \dots \right] (x^2 + 1) \\
&= -\frac{1}{6D} \left[1 + \frac{D^2}{6} + \left(\frac{D}{6} \right)^2 - \frac{D}{6} \dots \right] (x^2 + 1) \\
&= -\frac{1}{6D} \left[x^2 + 1 + \frac{1}{3} - \frac{2x}{6} + \frac{1}{18} \right] \\
&= -\frac{1}{6D} \left[x^2 - \frac{x}{3} + \frac{25}{18} \right]
\end{aligned}$$

$$= -\frac{1}{6} \left[\frac{x^3}{3} - \frac{x^2}{6} + \frac{25x}{18} \right]$$

$$y = A + Be^{-2x} + Ce^{3x} - \frac{1}{6} \left[\frac{x^3}{3} - \frac{x^2}{6} + \frac{25x}{18} \right]$$

RULE 5:

When $X = e^{ax}V$,

Where $V = \sin ax$ (or) $\cos ax$ (or) x^n

$$PI = \frac{1}{f(D)} e^{ax} V = e^{ax} \frac{1}{f(D+a)} V$$

PROBLEMS

1. $(D^2 + 4D + 4)y = e^{-x} \sin 2x$

Solution

$$m^2 + 4m + 4 = 0$$

$$(m + 2)^2 = 0$$

$$m = -2(\text{twice})$$

$$y = CF = (Ax + B)e^{-2x}$$

$$PI = \frac{1}{D^2 + 4D + 4} e^{-x} \sin 2x$$

$$= e^{-x} \frac{1}{(D-1)^2 + 4(D-1) + 4} \sin 2x$$

$$= e^{-x} \frac{1}{D^2 + 2D + 1} \sin 2x$$

$$= e^{-x} \frac{1}{-4 + 2D + 1} \sin 2x$$

$$= e^{-x} \frac{1}{2D - 3} \sin 2x$$

$$= e^{-x} \frac{2D + 3}{4D^2 - 9} \sin 2x$$

$$= e^{-x} \frac{2D + 3}{-25} \sin 2x$$

$$= e^{-x} \frac{1}{-25} [2D(\sin 2x) + 3 \sin 2x]$$

$$= e^{-x} \frac{1}{-25} [4 \cos 2x + 3 \sin 2x]$$

$$y = (Ax + B)e^{-2x} + e^{-x} \frac{1}{-25} [4 \cos 2x + 3 \sin 2x]$$

$$2.(D^2 + 9)y = (x^2 + 1)e^{3x}$$

Solution

The auxillary equation is

$$m^2 + 9 = 0$$

$$m = \pm 3i$$

$$y = CF = A \cos 3x + B \sin 3x$$

$$PI = \frac{1}{D^2 + 9} (x^2 + 1)e^{3x}$$

$$= e^{3x} \frac{1}{(D + 3)^2 + 9} (x^2 + 1)$$

$$= e^{3x} \frac{1}{(D)^2 + 6D + 18} (x^2 + 1)$$

$$= e^{3x} \frac{1}{18[1 + \frac{(D)^2 + 6D}{18}]} (x^2 + 1)$$

$$= e^{3x} \frac{1}{18} [1 + \frac{(D)^2 + 6D}{18}]^{-1} (x^2 + 1)$$

$$= \frac{e^{3x}}{18} [1 - \frac{(D)^2 + 6D}{18} + \left(\frac{(D)^2 + 6D}{18}\right)^2 - \dots] (x^2 + 1)$$

$$= \frac{e^{3x}}{18} [1 - \frac{(D)^2}{18} - \frac{D}{3} + \dots] (x^2 + 1)$$

$$= \frac{e^{3x}}{18} [x^2 + 1 - \frac{2}{18} - \frac{2x}{3} + \frac{2}{9}]$$

$$= \frac{e^{3x}}{18} [x^2 - \frac{2x}{3} + \frac{10}{9}]$$

$$y = A \cos 3x + B \sin 3x + \frac{e^{3x}}{18} [x^2 - \frac{2x}{3} + \frac{10}{9}]$$

RULE 6:

When $f(x) = x^n \sin ax$ (or) $x^n \cos ax$

Particular Integral

$$\frac{1}{f(D)} x^n \sin ax \text{ (or) } x^n \cos ax$$

$$= \frac{1}{f(D)} x^n (\cos ax + i \sin ax) = \frac{1}{f(D)} x^n e^{iax}$$

$$= e^{iax} \frac{1}{f(D + ia)} x^n$$

$$\frac{1}{f(D)} x^n \sin ax = \text{Im}g e^{iax} \frac{1}{f(D + ia)} x^n$$

$$\frac{1}{f(D)} x^n \cos ax = \text{Real } e^{iax} \frac{1}{f(D + ia)} x^n$$

PROBLEMS:

1. $(D^2 - 2D + 1)y = x \sin x$

Solution

$$m^2 - 2m + 1 = 0$$

$$(m - 1)^2 = 0$$

$$m = 1(\text{twice})$$

$$CF = y = (Ax + B)e^x$$

$$PI = \frac{1}{D^2 - 2D + 1} x \sin x$$

$$= \text{Im}g \frac{1}{D^2 - 2D + 1} x (\cos x + i \sin x)$$

$$= \text{Im}g \frac{1}{D^2 - 2D + 1} x (e^{ix})$$

$$= \text{Im}g(e^{ix}) \frac{1}{(D + i)^2 - 2(D + i) + 1} x$$

$$= \text{Im}g(e^{ix}) \frac{1}{-2i \left[1 + \frac{D^2 - 2(1 - i)D}{-2i} \right]} x$$

$$= \text{Im}g(e^{ix}) \frac{1}{-2i} \left[1 + \frac{D^2 - 2(1 - i)D}{-2i} \right]^{-1} x$$

$$= \text{Im}gi \frac{e^{ix}}{2} \left[1 + \frac{D^2 - 2(1 - i)D}{-2i} + \dots \right] x$$

$$= \text{Im}gi \frac{e^{ix}}{2} \left[1 + \frac{D^2}{2i}(x) - \frac{2(1 - i)D(x)}{i} \right]$$

$$\begin{aligned}
&= \operatorname{Im} gi \frac{e^{ix}}{2} \left[1 - \frac{(1-i)D(x)}{i} \right] \\
&= \operatorname{Im} gi \frac{e^{ix}}{2} \left[1 - \frac{(1-i)D(x)}{i} \right] \\
&= \operatorname{Im} gi \frac{e^{ix}}{2} [1 + i(1-i)D(x)] \\
&= \operatorname{Im} gi \frac{e^{ix}}{2} [1 + (i+1)D(x)] \\
&= \operatorname{Im} gi \frac{e^{ix}}{2} [1 + (i+1)] \\
&= \operatorname{Im} gi \frac{e^{ix}}{2} [x + (i+1)] \\
&= \operatorname{Im} gi \frac{(\cos x + i \sin x)}{2} [x + (i+1)] \\
&= \operatorname{Im} g \frac{(i \cos x - \sin x)}{2} [x + (i+1)] \\
&= \frac{1}{2} [x \cos x + \cos x - \sin x]
\end{aligned}$$

2. $(D^2 + D)y = x \cos x$

Solution

The auxillary equation is

$$\begin{aligned}
m^2 + m &= 0 \\
m(m+1) &= 0 \\
m = 0, m &= -1 \\
CF &= Ae^{0x} + Be^{-x} \\
y &= \frac{1}{D^2 + D} x \cos x \\
&= R.P \text{ of } \frac{1}{D^2 + D} x e^{ix} \\
&= R.P \text{ of } e^{ix} \frac{1}{(D+i)^2 + (D+i)} x \\
&= R.P \text{ of } e^{ix} \frac{1}{(D)^2 - 1 + 2iD + D + i} x
\end{aligned}$$

$$\begin{aligned}
&= R.P \text{ of } e^{ix} \frac{1}{(D)^2 + (1 + 2i)D + (1 + i)} x \\
&= R.P \text{ of } e^{ix} \frac{1}{(i - 1)(1 + \frac{(1 + 2i)D + D^2}{i - 1})} x \\
&= R.P \text{ of } e^{ix} \frac{1}{(i - 1)} \left[1 + \frac{(1 + 2i)D + D^2}{i - 1} \right]^{-1} x \\
&= R.P \text{ of } e^{ix} \frac{1}{(i - 1)} \left[1 - \frac{(1 + 2i)D + D^2}{i - 1} \right] x \\
&= R.P \text{ of } e^{ix} \frac{1}{(i - 1)} \left[x - \frac{(1 + 2i)D(x)}{i - 1} \right] \\
&= R.P \text{ of } e^{ix} \frac{1}{(i - 1)} \left[x - \frac{(1 + 2i)}{i - 1} \right] \\
&= R.P \text{ of } e^{ix} \left[\frac{x}{i - 1} - \frac{1 - 2i}{(i - 1)^2} \right] \\
&= R.P \text{ of } e^{ix} \left[\frac{x(i + 1)}{-2} + \frac{1(-i + 2)}{2} \right] \\
&= R.P \text{ of } \frac{\cos x + i \sin x}{2} [-(xi + x) - i + 2] \\
&= R.P \text{ of } \frac{\cos x + i \sin x}{2} [2 - x - i(x + 1)] \\
&= \frac{1}{2} [(2 - x) \cos x + (x + 1) \sin x] \\
&= \cos x + \frac{\sin x}{2} + \frac{x}{2} (\sin x - \cos x)
\end{aligned}$$

RULE 7:

When X is any other function of x $PI = \frac{1}{f(D)} X$

If $f(D) = (D - m_1)(D - m_2) \dots (D - m_n)$ after resolving into partial fractions

Then,

$$\begin{aligned}
\frac{1}{f(D)} &= \frac{A_1}{D - m_1} + \frac{A_2}{D - m_2} + \dots + \frac{A_n}{D - m_n} \\
PI &= \left[\frac{A_1}{D - m_1} + \frac{A_2}{D - m_2} + \dots + \frac{A_n}{D - m_n} \right] X \\
&= A_1 \left[\frac{1}{D - m_1} \right] X + A_2 \left[\frac{1}{D - m_2} \right] X + \dots + A_n \left[\frac{1}{D - m_n} \right] X
\end{aligned}$$

$$= A_1 e^{m_1 x} \int X e^{-m_1 x} dx + A_2 e^{m_2 x} \int X e^{-m_2 x} dx + \dots + A_n e^{m_n x} \int X e^{-m_n x} dx$$

PROBLEMS:

1.Solve $(D^2 - 4D + 13)y = \cos 3x. \cosh 2x$

Solution

The auxillary solution is

$$(m^2 - 4m + 13) = 0$$

$$m = \frac{4 \pm \sqrt{16 - 52}}{2} = \frac{4 \pm \sqrt{-36}}{2} = \frac{4 \pm 6i}{2} = 2 \pm 3i$$

$$CF = e^{2x}(A \cos 3x + B \sin 3x)$$

$$RHS = \cos 3x. \cosh 2x$$

$$= \cos 3x. \frac{(e^{2x} + e^{-2x})}{2}$$

$$= (e^{2x} \cos 3x + e^{-2x} \cos 3x). \frac{1}{2}$$

$$PI_1 = \frac{1}{D^2 - 4D + 13} e^{2x} \cos 3x$$

$$= e^{2x} \frac{1}{(D+2)^2 - 4(D+2) + 13} \cos 3x$$

$$= e^{2x} \frac{1}{(D)^2 + 4D + 4 - 4(D) - 8 + 13} \cos 3x$$

$$= e^{2x} \frac{1}{(D)^2 + 9} \cos 3x$$

$$= e^{2x} \frac{1}{-9 + 9} \cos 3x$$

$$= e^{2x} \frac{x}{2D} \cos 3x$$

$$= e^{2x} \frac{x \sin 3x}{2 \cdot 3}$$

$$= \frac{x e^{2x} \sin 3x}{6}$$

$$PI_2 = \frac{1}{D^2 - 4D + 13} e^{-2x} \cos 3x$$

$$= e^{-2x} \frac{1}{(D-2)^2 - 4(D-2) + 13} \cos 3x$$

$$\begin{aligned}
&= e^{-2x} \frac{1}{(D)^2 - 4D + 4 - 4(D) + 8 + 13} \cos 3x \\
&= e^{-2x} \frac{1}{(D)^2 - 8D + 25} \cos 3x \\
&= e^{-2x} \frac{1}{-9 - 8D + 25} \cos 3x \\
&= e^{-2x} \frac{1}{16 - 8D} \cos 3x \\
&= \frac{e^{-2x}}{8} \frac{(2 + D)}{(2 - D)(2 + D)} \cos 3x \\
&= \frac{e^{-2x}}{8} \frac{(2 + D)}{4 - D^2} \cos 3x \\
&= \frac{e^{-2x}}{8} \frac{(2 + D)}{4 - (-9)} \cos 3x \\
&= \frac{e^{-2x}}{8} \frac{(2 + D)}{13} \cos 3x \\
&= \frac{e^{-2x}}{104} [2 \cos 3x - 3 \sin 3x]
\end{aligned}$$

$$y = e^{2x}(A \cos 3x + B \sin 3x) + \frac{x e^{2x} \sin 3x}{6} + \frac{e^{-2x}}{104} [2 \cos 3x - 3 \sin 3x]$$

2. Solve $(D^2 + 2D + 1)y = x^3 + \sin 2x \cos x$

Solution

The auxillary equation is

$$\begin{aligned}
m^2 + 2m + 1 &= 0 \\
(m + 1)^2 &= 0 \\
m &= -1, -1 \\
CF &= (Ax + B)e^{-x} \\
PI_1 &= \frac{1}{(D + 1)^2} x^3 \\
&= (1 + D)^{-2} x^3 \\
&= (1 - 2D + 3D^2 - 4D^3 + \dots) x^3 \\
&= x^3 - 2(3x^2) + 3(6x) - 4(6) \\
&= x^3 - (6x^2) + (18x) - 24
\end{aligned}$$

$$\begin{aligned}
PI_2 &= \frac{1}{(D)^2 + 2D + 1} \sin 2x \cos x \\
&= \frac{1}{(D)^2 + 2D + 1} \left[\frac{1}{2} (\sin 3x + \sin x) \right] \\
&= \frac{1}{2} \left[\frac{1}{(D)^2 + 2D + 1} \sin 3x + \frac{1}{(D)^2 + 2D + 1} \sin x \right] \\
&= \frac{1}{2} \left[\frac{1}{-9 + 2D + 1} \sin 3x + \frac{1}{-1 + 2D + 1} \sin x \right] \\
&= \frac{1}{2} \left[\frac{1}{2D - 8} \sin 3x + \frac{1}{2D} \sin x \right] \\
&= \frac{1}{2} \left[\frac{2D + 8}{(2D - 8)(2D + 8)} \sin 3x - \frac{1}{2} \cos x \right] \\
&= \frac{1}{2} \left[\frac{2D + 8}{4(D)^2 - 64} \sin 3x - \frac{1}{2} \cos x \right] \\
&= \frac{1}{2} \left[\frac{2D + 8}{4(-9) - 64} \sin 3x - \frac{1}{2} \cos x \right] \\
&= \frac{1}{2} \left[\frac{-1}{100} [2(3 \cos 3x + 4 \sin 3x)] - \frac{1}{2} \cos x \right] \\
&= \frac{1}{4} \left[\frac{-1}{25} [(3 \cos 3x + 4 \sin 3x)] - \cos x \right] \\
&= \frac{\cos x}{4} - \frac{1}{100} (3 \cos 3x + 4 \sin 3x)
\end{aligned}$$

$$y = (Ax + B)e^{-x} + x^3 - (6x^2) + (18x) - 24 + \frac{\cos x}{4} - \frac{1}{100} (3 \cos 3x + 4 \sin 3x)$$

3.Solve $(D^2 - 2D + 1)y = x e^x \sin x$

Solution

The auxillary solution is

$$(m^2 - 2m + 1) = 0$$

$$(m - 1)^2 = 0$$

$$m = 1, 1$$

$$y = CF = (Ax + B)e^x$$

$$PI = \frac{1}{(D - 1)^2} x e^x \sin x$$

$$= e^x \frac{1}{((D+1)-1)^2} x \sin x$$

$$= e^x \frac{1}{(D)^2} x \sin x$$

$$= e^x \left[x - \frac{2D}{(D)^2} \right] \frac{1}{(D)^2} \sin x$$

$$= e^x \left[x - \frac{2}{D} \right] (-\sin x)$$

$$= e^x [-x \sin x - 2 \cos x]$$

$$= -e^x [x \sin x + 2 \cos x]$$

$$y = (Ax + B)e^x - e^x [x \sin x + 2 \cos x]$$

PRACTICE PROBLEMS

Solve

$$1. (D^2 + 4)y = x^4 + \cos^2 x$$

$$2. (D^2 + 5D + 4)y = e^{-x} \sin 2x$$

$$3. (D^3 - 3D^2 + 4D - 2)y = e^x + \cos x$$

$$4. (D^2 + 2D + 3)y = 5x^2$$

$$5. (D^3 - 7D - 6)y = (1 + x)e^{2x}$$

$$6. (D^2 - 1)y = x^2 \cos x$$

$$7. (D^4 + a^4)y = \cos ax$$

$$8. (D^2 - 3D + 2)y = 1 + x$$

$$9. (D^3 - 5D^2 + 7D - 3)y = e^{2x} \cosh x$$

$$10. (D^2 + 2D - 1)y = (x + e^x)^2$$

EULER'S AND LEGENDER'S EQUATION

EULER'S EQUATION

The differential equation

$$a_0 x^n \frac{d^n y}{dx^n} + a_1 x^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + a_2 x^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + a_3 x^{n-3} \frac{d^{n-3} y}{dx^{n-3}} + \dots + a_{n-1} x \frac{dy}{dx} + a_n y = X$$

Where $a_0, a_1, a_2, \dots, a_n$ are constants and X is a function of 'x' is called a homogeneous linear equation of order 'n' with variable coefficients. This equation is called **Euler's equation (or) Cauchy's equation**.

To solve this type of equation let us reduce this to differential equations with constant coefficients

Procedure

Put $Z = \log x$ (or) $x = e^z$

$$\text{As } \frac{dy}{dx} = \frac{dy}{dz} \frac{dz}{dx} = \frac{dy}{dz} \frac{1}{x} \text{ (Since } z = \log x \text{)}$$

$$\text{So } x \frac{dy}{dx} = \frac{dy}{dz} = \theta y \left(\begin{array}{l} \text{Since } \frac{d}{dz} = \theta \\ \frac{d}{dx} = D \end{array} \right)$$

$$\text{Similarly } x^2 \frac{d^2 y}{dx^2} = \theta(\theta - 1)y$$

$$x^3 \frac{d^3 y}{dx^3} = \theta(\theta - 1)(\theta - 2)y$$

$$\dots\dots\dots x^n \frac{d^n y}{dx^n} = \theta(\theta - 1)(\theta - 2) \dots \dots (\theta - (n - 1))y$$

Example 1

$$\text{Solve } x^2 \frac{d^2 y}{dx^2} - 7x \frac{dy}{dx} + 12y = x^2 \log x + x^3$$

Solution:

The given equation is an **Euler's equation**

Put $Z = \log x$ (or) $x = e^z$

$$x \frac{dy}{dx} = \theta y$$

$$x^2 \frac{d^2 y}{dx^2} = \theta(\theta - 1)y$$

Hence the given equation becomes as,

$$\theta(\theta - 1)y - 7\theta y + 12y = e^{2z}z + e^{3z}$$

$$\theta^2 y - 8\theta y + 12y = e^{2z}z + e^{3z}$$

$$[\theta^2 - 8\theta + 12]y = e^{2z}z + e^{3z}$$

This a differential equation with constant coefficients.

Hence to find the Complementary function and Particular Integral for the above differential equation.

Complementary Function(C.F)

The auxiliary equation is $m^2 - 8m + 12 = 0$; $m = 6, 2$ are the roots.

Hence C.F = $Ae^{6z} + Be^{2z}$

Particular Integral(P.I)

To find P.I₁ and P.I₂ as there are two functions on the R.H.S

$$P.I_1 = \frac{1}{[\theta^2 - 8\theta + 12]} e^{2z} z$$

By Exponential Shift Rule replace ' θ ' by ' $\theta + 2$ ' then,

$$P.I_1 = e^{2z} \frac{1}{[(\theta+2)^2 - 8(\theta+2) + 12]} z$$

$$P.I_1 = e^{2z} \frac{1}{[(\theta^2 + 4 + 4\theta - 8\theta - 16 + 12)]} z$$

$$P.I_1 = e^{2z} \frac{1}{[(\theta^2 - 4\theta)]} z = e^{2z} \frac{1}{[4\theta(1 - \frac{\theta}{4})]} z$$

$$= \frac{e^{2z}}{4\theta} \left[\left(1 - \frac{\theta}{4}\right)^{(-1)} \right] z$$

$$= \frac{e^{2z}}{4\theta} \left[1 + \frac{\theta}{4} + \left(\frac{\theta}{4}\right)^2 + \left(\frac{\theta}{4}\right)^3 + \dots + \left(\frac{\theta}{4}\right)^n \right] z$$

$$= \frac{e^{2z}}{4\theta} \left[z + \frac{\theta \cdot z}{4} \right] \text{ (After omitting higher powers of derivatives)}$$

$$= \frac{e^{2z}}{4\theta} \left[z + \frac{1}{4} \right] = \frac{e^{2z}}{4} \frac{1}{\theta} \left[z + \frac{1}{4} \right] = \frac{e^{2z}}{4} \left[\frac{z^2}{2} + \frac{1}{4} z \right]$$

As $\frac{1}{\theta}$ Integrating with respect to variable ' z '.

$$P.I_2 = \frac{1}{[\theta^2 - 8\theta + 12]} e^{3z}$$

By Rule 1, $X = e^{ax}$, where a is a constant replace ' θ ' by ' 3 '

$$P.I_2 = \frac{1}{[3^2 - 8(3) + 12]} e^{3z} = \frac{e^{3z}}{-3}$$

Hence the solution is $Z = C.F + P.I$ (P.I₁ + P.I₂)

$$\text{So } Z = Ae^{6z} + Be^{2z} + \frac{e^{2z}}{4} \left[\frac{z^2}{2} + \frac{1}{4} z \right] + \frac{e^{3z}}{-3}$$

Example 2:

$$\text{Solve } (x^2 D^2 + xD - 9)y = \frac{5}{x^2}$$

Solution: The given equation is an **Euler's equation**

Put $Z = \log x$ (or) $x = e^z$

$$x \frac{dy}{dx} = \theta y$$

$$x^2 \frac{d^2 y}{dx^2} = \theta(\theta - 1)y$$

So given equation becomes as,

$$(\theta(\theta - 1) + \theta - 9)y = 5e^{-2z}$$

$$(\theta^2 - 9)y = 5e^{-2z}$$

Hence to find the Complementary function and Particular Integral for the above differential equations.

Complementary Function(C.F)

The auxiliary equation is $m^2 - 9 = 0$; $m = \pm 3$ are the roots.

$$\text{Hence C.F} = Ae^{3z} + Be^{-3z}$$

Particular Integral(P.I)

$$\text{P.I} = \frac{1}{\theta^2 - 9} 5e^{-2z}$$

By Rule 1, $X = e^{ax}$, where a is a constant Replace ' θ ' by ' -2 '

$$\text{P.I}_2 = \frac{1}{[(-2)^2 - 9]} 5e^{-2z} = 5 \frac{e^{-2z}}{-5} = -e^{-2z}$$

Hence the solution is $Z = \text{C.F} + \text{P.I}$

$$\text{So } Z = Ae^{3z} + Be^{-3z} - e^{-2z}$$

$$Z = Ae^{3 \log x} + Be^{-3 \log x} - e^{-2 \log x}$$

$$Z = Ae^{\log x^3} + Be^{\log x^{-3}} - e^{\log x^{-2}}$$

$$Z = Ax^3 + Bx^{-3} - x^{-2}$$

Example 3:

Solve $\left(\frac{d^2y}{dx^2} + \frac{1}{x} \frac{dy}{dx}\right) = \frac{12\log x}{x^2}$

Solution: The given equation is an **Euler's equation**

The given equation is not in homogeneous form.

So multiply by 'x²', then $\left(x^2 \frac{d^2y}{dx^2} + x \frac{dy}{dx}\right) = 12\log x$

Put Z= logx (or) x=e^z

$$x \frac{dy}{dx} = \theta y$$

$$x^2 \frac{d^2y}{dx^2} = \theta(\theta - 1)y$$

So given equation becomes as,

$$(\theta(\theta - 1) + \theta)y = 12z$$

$$(\theta^2)y = 12z$$

Hence to find the Complementary function and Particular Integral for the above differential equations.

Complementary Function(C.F)

The auxiliary equation is $m^2 = 0$; $m = 0,0$ are the roots.

$$\text{Hence C.F} = (Az + B)e^{0z} = Az + B$$

Particular Integral(P.I)

$$\text{P.I} = \frac{1}{\theta^2} 12z$$

By Rule 3,

$$\text{P.I} = \frac{1}{[\theta]} 12 \frac{z^2}{2} = \frac{1}{[\theta]} 6z^2 = 2z^3$$

Hence the solution is Z= C.F+P.I

$$Z = Az + B + 2z^3$$

$$Z = A\log x + B + 2(\log x)^3$$

Example 4:

Solve $\left(x^3 \frac{d^3y}{dx^3} + 2x^2 \frac{d^2y}{dx^2} + 2y\right) = 10\left(x + \frac{1}{x}\right)$

Solution: The given equation is an **Euler's equation**

Given equation to multiply by 'x²', then $\left(x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx}\right) = 12 \log x$

Put $Z = \log x$ (or) $x = e^z$

$$x \frac{dy}{dx} = \theta y$$

$$x^2 \frac{d^2 y}{dx^2} = \theta(\theta - 1)y$$

$$x^3 \frac{d^3 y}{dx^3} = \theta(\theta - 1)(\theta - 2)y$$

So given equation becomes as,

$$(\theta(\theta - 1)(\theta - 2) + 2\theta(\theta - 1)y + 2\theta)y = 10(e^z + e^{-z})$$

$$(\theta^3 - \theta^2 + 2)y = 10(e^z + e^{-z})$$

Hence to find the Complementary function and Particular Integral for the above differential equations.

Complementary Function(C.F)

The auxiliary equation is $(m^3 - m^2 + 2) = 0$; $m = -1, 1 \pm i$ are the roots.

Hence C.F = $Ae^{-z} + e^z(BCosz + CSinz)$

Particular Integral(P.I)

$$P.I = \frac{1}{\theta^3 - \theta^2 + 2} 10(e^z + e^{-z})$$

By Rule 1 $X = e^{ax}$, where a is a constant Replace ' θ ' by ' -1 '

$$\begin{aligned} P.I &= \frac{10e^z}{[\theta^3 - \theta^2 + 2]} + \frac{10e^{-z}}{[\theta^3 - \theta^2 + 2]} = \frac{10e^z}{[1^3 - 1^2 + 2]} + \frac{10e^{-z}}{[(-1)^3 - (-1)^2 + 2]} = \frac{10e^z}{2} + \frac{10e^{-z}}{0} \\ &= 5e^z + \frac{10ze^{-z}}{(3D^2 - 2D)} \end{aligned}$$

(As Denominator get's zero multiply the numerator by 'z' and differentiate the denominator with respect to 'D')

$$P.I = 5e^z + \frac{10ze^{-z}}{3(-1)^2 - 2(-1)} = 5e^z + \frac{10ze^{-z}}{5}$$

$$P.I = 5e^z + 2ze^{-z}$$

Hence the solution is $Z = C.F + P.I$

$$Z = Ae^{-z} + e^z(BCosz + CSinz) + 5e^z + 2ze^{-z}$$

$$Z = Ae^{-\log x} + e^{\log x}(BCos\log x + CSin\log x) + 5e^{\log x} + 2\log x e^{-\log x}$$

$$Z = \frac{A}{x} + x(BCos\log x + CSin\log x) + 5x + \frac{2\log x}{x}$$

Example 5:

$$\text{Solve } (x^2 \frac{d^2 y}{dx^2} + 4x \frac{dy}{dx} + 2y) = (x^2 + \frac{1}{x^2})$$

Solution: The given equation is an **Euler's equation**

$$\text{Given equation is } (x^2 D^2 + 4xD + 2)y = \left(x^2 + \frac{1}{x^2}\right) \text{ where } D = \frac{d}{dx}$$

Put $Z = \log x$ (or) $x = e^z$

$$x \frac{dy}{dx} = \theta y$$

$$x^2 \frac{d^2 y}{dx^2} = \theta(\theta - 1)y$$

So given equation becomes as,

$$(\theta(\theta - 1) + 4\theta + 2)y = (e^{2z} + e^{-2z})$$

$$(\theta^2 + 3\theta + 2)y = (e^{2z} + e^{-2z})$$

Hence to find the Complementary function and Particular Integral for the above differential equations.

Complementary Function(C.F)

The auxiliary equation is $(m^2 + 3m + 2) = 0$; $m = -1, -2$ are the roots.

$$\text{Hence C.F} = Ae^{-z} + Be^{-2z}$$

Particular Integral(P.I)

$$P.I = \frac{1}{\theta^2 + 3\theta + 2} (e^{2z} + e^{-2z})$$

By Rule 1 $X = e^{ax}$, where a is a constant Replace ' θ ' by ' 2 '

$$P.I = \frac{1}{\theta^2+3\theta+2} e^{2z} + \frac{1}{\theta^2+3\theta+2} e^{-2z} = \frac{1e^{2z}}{[2^2+6+2]} + \frac{1e^{-2z}}{[(-2)^2-6+2]} = \frac{e^{2z}}{12} + \frac{e^{-2z}}{0}$$

$$= \frac{e^{2z}}{12} + \frac{ze^{-2z}}{(2D+3)}$$

(As Denominator get's zero multiply the numerator by 'z' and differentiate the denominator with respect to 'D')

$$P.I = \frac{e^{2z}}{12} + \frac{ze^{-2z}}{(2(-2)+3)} = \frac{e^{2z}}{12} - ze^{-2z}$$

$$P.I = \frac{e^{2z}}{12} - ze^{-2z} \text{ Hence the solution is } Z = C.F + P.I$$

$$Z = Ae^{-z} + Be^{-2z} + \frac{e^{2z}}{12} - ze^{-2z}$$

$$Z = Ae^{-\log x} + Be^{-2\log x} + \frac{e^{2\log x}}{12} - \log x e^{-2\log x}$$

$$Z = \frac{A}{x} + \frac{B}{x^2} + \frac{x^2}{12} - \frac{\log x}{x^2}$$

Example 6:

$$\text{Solve } (x^2 \frac{d^2 y}{dx^2} - 2x \frac{dy}{dx} - 4y) = 32(\log x)^2$$

Solution: The given equation is an **Euler's equation**

$$\text{Given equation is } (x^2 D^2 - 2xD - 4)y = 32((\log x)^2) \text{ where } D = \frac{d}{dx}$$

$$\text{Put } Z = \log x \text{ (or) } x = e^z$$

$$x \frac{dy}{dx} = \theta y$$

$$x^2 \frac{d^2 y}{dx^2} = \theta(\theta - 1)y$$

So given equation becomes as,

$$(\theta(\theta - 1) - 2\theta - 4)y = 32z^2$$

$(\theta^2 - 3\theta - 4)y = 32z^2$ Hence to find the Complementary function and Particular Integral for the above differential equations.

Complementary Function(C.F)s

The auxiliary equation is $(m^2 - 3m - 4) = 0$; $m = -1, 4$ are the roots.

$$\text{Hence C.F} = Ae^{-z} + Be^{4z}$$

Particular Integral(P.I)

$$P.I = \frac{1}{\theta^2 - 3\theta - 4} 32z^2$$

By Rule 3 where X expanding by using Binomial expansion,

$$P.I = \frac{1}{\theta^2 - 3\theta - 4} 32z^2 = \frac{32}{-4} \left[\frac{1}{\left[1 - \left(\frac{\theta^2 - 3\theta}{4}\right)\right]} z^2 \right] = -8 \left[\left(1 - \left(\frac{\theta^2 - 3\theta}{4}\right)\right)^{(-1)} \right] z^2$$

$$= -8 \left[1 + \frac{\theta(\theta - 3)}{4} + \frac{\theta^2(\theta - 3)^2}{16} \right] z^2 \quad (\text{After omitting higher powers of } \theta)$$

$$P.I = -8 \left[z^2 - \frac{3\theta(z^2)}{4} + \frac{13}{16} \theta^2(z^2) \right]$$

$$P.I = -8 \left[z^2 - \frac{3(2z)}{4} + \frac{13}{8} \right] \quad \text{Hence the solution is } Z = C.F + P.I$$

$$Z = Ae^{-\log x} + Be^{4\log x} + -8 \left[(\log x)^2 - \frac{3(2\log x)}{4} + \frac{13}{8} \right]$$

$$Z = \frac{A}{x} + \frac{B}{x^4} + [-8(\log x)^2 + 12\log x - 13]$$

LEGENDER'S EQUATION

The differential equation

$$b_0(px + q)^n \frac{d^n y}{dx^n} + b_1(px + q)^{n-1} \frac{d^{n-1} y}{dx^{n-1}} + b_2(px + q)^{n-2} \frac{d^{n-2} y}{dx^{n-2}} + \dots + b_n(px + q)^0 \frac{d^0 y}{dx^0} = X$$

Where $b_0, b_1, b_2, \dots, b_n$ are constants and p & q are constants and ' x ' is a function of ' x ' is called a homogeneous linear equation of order ' n ' with variable coefficients. This equation is called **LEGENDER'S LINEAR EQUATION**

To solve this type of equation let us reduce this to differential equations with constant coefficients

Procedure

Put $Z = \log(px + q)$ (or) $px + q = e^z$

$$x = \frac{e^z - q}{p}$$

$$\text{Similarly as Euler's equation } (px + q) \frac{dy}{dx} = p\theta y \left(\begin{array}{l} \text{Since } \frac{d}{dz} = \theta \\ \frac{d}{dx} = D \end{array} \right)$$

$$\text{Similarly } (px + q)^2 \frac{d^2y}{dx^2} = p^2\theta(\theta - 1)y$$

$$(px + q)^3 \frac{d^3y}{dx^3} = p^3\theta(\theta - 1)(\theta - 2)y$$

$$\dots\dots\dots (px + q)^n \frac{d^ny}{dx^n} = p^n\theta(\theta - 1)(\theta - 2) \dots\dots (\theta - (n - 1))y$$

Example 1:

$$\text{Solve } (2x - 1)^2 D^2 - 4(2x - 1)D + 8)y = 8x$$

Solution: The given equation is a Legendre's Equation.

$$\text{Put } Z = \log(2x-1) \text{ (or) } 2x-1=e^z$$

$$X = \frac{e^z + 1}{2}$$

$$(2x - 1) \frac{dy}{dx} = 2\theta y$$

$$(2x - 1)^2 \frac{d^2y}{dx^2} = 4\theta(\theta - 1)y$$

So given equation becomes as,

$$(4\theta(\theta - 1) - 4(2\theta) + 8)y = 8\left(\frac{e^z + 1}{2}\right)$$

$$(4\theta^2 - 12\theta + 8)y = 4(e^z + 1)$$

$$4(\theta^2 - 3\theta + 2)y = 4(e^z + 1)$$

$$(\theta^2 - 3\theta + 2)y = (e^z + 1)$$

Hence to find the Complementary function and Particular Integral for the above differential equations.

Complementary Function(C.F)

The auxiliary equation is $m^2 - 3m + 2 = 0$; $m = 1, 2$ are the roots.

$$\text{Hence C.F} = Ae^z + Be^{2z}$$

Particular Integral(P.I)

$$P.I = \frac{1}{\theta^2 - 3\theta + 2} (e^z + 1)$$

By Rule 1, Replace ' θ ' by '1'

$$P.I = \frac{1}{1^2 - 3(1) + 2} (e^z) + \frac{1}{0^2 - 3(0) + 2} e^{0z} = \frac{1}{0} (e^z) + \frac{1}{2} = \frac{ze^z}{2(1) - 3} + \frac{1}{2}$$

(As Denominator get's zero multiply the numerator by 'z' and differentiate the denominator with respect to 'D')

$$P.I = -ze^z + \frac{1}{2}$$

Hence the solution is $Z = C.F + P.I$

$$\text{So } Z = Ae^{\log(2x-1)} + Be^{2\log(2x-1)} - \log(2x-1) e^{\log(2x-1)} + \frac{1}{2}$$

$$Z = A(2x-1) + B(2x-1)^2 - (2x-1) \log(2x-1) + \frac{1}{2}$$

Example 2:

$$\text{Solve } \left((x+1)^2 \frac{d^2}{dx^2} + (x+1) \frac{d}{dx} + 1 \right) y = 4 \cos \log(1+x)$$

Solution: The given equation is a Legendre's Equation.

$$\text{Put } Z = \log(x+1) \text{ (or) } x+1 = e^z$$

$$X = e^z - 1$$

$$(x+1) \frac{dy}{dx} = \theta y$$

$$(x+1)^2 \frac{d^2 y}{dx^2} = \theta(\theta-1)y$$

So given equation becomes as,

$$(\theta(\theta-1) + \theta + 1)y = 4 \cos z$$

$$(\theta^2 + 1)y = 4 \cos z$$

Hence to find the Complementary function and Particular Integral for the above differential equations.

Complementary Function(C.F)

The auxiliary equation is $m^2 + 1 = 0$; $m = \pm i$ are the roots.

Hence C.F = $ACosz + BSinz$

Particular Integral(P.I)

$$P.I = \frac{1}{\theta^2 + 1} 4Cosz$$

By Rule 2 where X is a trigonometric functions, Replace ' θ^2 ' by '-1'

$$P.I = \frac{1}{-1+1} (4Cosz) = \frac{1}{0} (4Cosz) = \frac{4zCosz}{2\theta}$$

(As Denominator get's zero multiply the numerator by 'z' and differentiate the denominator with respect to 'D')

$$P.I = 2z(Sinz) + c$$

Hence the solution is $Z = C.F + P.I$

$$\text{So } Z = ACosz + BSinz + 2z(Sinz)$$

$$Z = ACos(\log(x + 1)) + B\sin(\log(x + 1)) + 2\log(x + 1)(\sin \log(x + 1))$$

Example 3:

$$\text{Solve } (2x + 3)^2 \frac{d^2y}{dx^2} - 2(2x + 3) \frac{dy}{dx} - 12y = 6x$$

Solution: The given equation is a Legendre's Equation.

$$\text{Put } Z = \log(2x+3) \text{ (or) } 2x+3=e^z$$

$$X = \frac{e^z - 3}{2}$$

$$(2x + 3) \frac{dy}{dx} = 2\theta y$$

$$(2x + 3)^2 \frac{d^2y}{dx^2} = 4\theta(\theta - 1)y$$

So given equation becomes as,

$$(4\theta(\theta - 1) - 2(2\theta) - 12)y = 6\left(\frac{e^z - 3}{2}\right)$$

$$(4\theta^2 - 8\theta - 12)y = 3(e^z - 3)$$

$$4(\theta^2 - 2\theta - 3)y = 3(e^z - 3)$$

$$(\theta^2 - 2\theta - 3)y = \frac{3}{4}(e^z - 3)$$

Hence to find the Complementary function and Particular Integral for the above differential equations.

Complementary Function(C.F)

The auxiliary equation is $m^2 - 2m - 3 = 0$; $m = 3, -1$ are the roots.

$$\text{Hence C.F} = Ae^{3z} + Be^{-z}$$

Particular Integral(P.I)

$$\text{P.I} = \frac{1}{(\theta^2 - 2\theta - 3)} \frac{3}{4}(e^z - 3) \text{ By Rule 1 } X = e^{ax}, \text{ where } a \text{ is a constant}$$

$$\text{P.I} = \frac{3}{4} \left[\frac{1}{(\theta^2 - 2\theta - 3)} (e^z) - \frac{1}{(\theta^2 - 2\theta - 3)} 3e^{0z} \right]$$

$$= \frac{3}{4} \left[\frac{1}{(1^2 - 2(1) - 3)} (e^z) - \frac{1}{(0^2 - 2(0) - 3)} 3e^{0z} \right] \text{ \{Replace '\theta' by '1' and Replace '\theta' by '0'\}}$$

$$= \frac{3}{4} \left[\frac{1}{-4} (e^z) - \frac{1}{(-3)} 3e^{0z} \right] = \frac{-3}{16} e^z + \frac{3}{4}$$

$$\text{P.I} = \frac{-3}{16} e^z + \frac{3}{4}$$

Hence the solution is $Z = C.F + P.I$

$$\text{So } Z = Ae^{3 \log(2x+3)} + Be^{-\log(2x+3)} + \frac{-3}{16} e^{\log(2x+3)} + \frac{3}{4}$$

$$Z = A(2x+3)^3 + \frac{B}{(2x+3)} - \frac{3}{16} (2x+3) + \frac{3}{4}$$

Method of variation of parameters

Introduction:

In many engineering heat transfer applications, the energy equation reduces to a second order differential equation with a nonlinear forcing function that is a function of both the independent variable and the dependent variable and its derivatives. Specifically, application of an energy balance to a system frequently results in a nonlinear differential equation that governs the temperature field in the system. A common method of obtaining analytical solutions to second order, nonhomogeneous differential equations is the method of variation of parameters in which the particular solution is assumed to be the sum of the products of unknown variable parameters and functions that make up a fundamental solution set to the corresponding homogeneous equation.

Consider the second order non-homogeneous equation: $\frac{d^2y}{dx^2} + P \frac{dy}{dx} + Qy = X$.

Let $y=f(x)$ and $y=g(x)$ be two independent solutions of the homogeneous equation $\frac{d^2y}{dx^2} + P \frac{dy}{dx} + Qy = 0$. Then $y = Af(x) + Bg(x)$, where A and B are arbitrary constants is the general solution. The general solution of the non-homogeneous equation: $\frac{d^2y}{dx^2} + P \frac{dy}{dx} + Qy = X$ takes the form $y = Af(x) + Bg(x)$, where A and B are not constants but functions of x.

The Wronskian of the functions $f(x)$ and $g(x)$ is given by $w = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix}$.

The values of $A(x)$ and $B(x)$ are obtained as follows:

$$A(x) = - \int \frac{Xg}{w} dx + C_1, \quad B(x) = \int \frac{Xf}{w} dx + C_2$$

Substituting for A(x) and B(x) in

$y = A(x)f(x) + B(x)g(x)$, the general solution of $\frac{d^2y}{dx^2} + P \frac{dy}{dx} + Qy = X$ is obtained.

Example 1- Solve: $\frac{d^2y}{dx^2} + y = \tan(x)$ by method of variation of parameters.

Solution: A.E is $m^2 + 1 = 0$

$$m^2 = \pm i$$

$$y = A \cos(x) + B \sin(x) \text{ -----(1)}$$

$$\text{Let } f=\cos(x) \text{ and } g=\sin(x), w = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} \cos(x) & \sin(x) \\ -\sin(x) & \cos(x) \end{vmatrix} = 1, X=\tan(x)$$

$$\begin{aligned} A &= -\int \frac{Xg}{w} dx + C_1, A = -\int \frac{\tan(x) \sin(x)}{1} dx = -\int \frac{\sin^2(x)}{\cos(x)} dx \\ &= -\int \frac{1-\cos^2(x)}{\cos(x)} dx = -\int \sec(x) dx + \int \cos(x) dx \\ &= -\log(\sec(x) + \tan(x)) + \sin(x) + C_1 \text{ -----(2)} \end{aligned}$$

$$B = \int \frac{Xf}{w} dx + C_2, B = \int \frac{\tan(x) \cos(x)}{1} dx = \int \sin(x) dx = -\cos(x) + C_2 \text{ -----(3)}$$

Substitute (2) and (3) in (1),

$$y = (-\log(\sec(x) + \tan(x)) + \sin(x) + C_1) \cos(x) + (-\cos(x) + C_2) \sin(x)$$

$$y = C_1 \cos(x) + C_2 \sin(x) - \cos(x) \log(\sec(x) + \tan(x))$$

Example 2- Solve: $\frac{d^2y}{dx^2} + a^2y = \tan(ax)$ by method of variation of parameters.

Solution: A.E is $m^2 + a^2 = 0$

$$m^2 = \pm ai$$

$$y = A \cos(ax) + B \sin(ax) \text{ -----(1)}$$

$$\text{Let } f=\cos(ax) \text{ and } g=\sin(ax), w = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} \cos(ax) & \sin(ax) \\ -a\sin(ax) & a\cos(ax) \end{vmatrix} = a,$$

$$X=\tan(ax)$$

$$\begin{aligned} A &= -\int \frac{Xg}{w} dx + C_1, A = -\int \frac{\tan(ax) \sin(ax)}{a} dx = -\frac{1}{a} \int \frac{\sin^2(ax)}{\cos(ax)} dx \\ &= -\frac{1}{a} \int \frac{1-\cos^2(ax)}{\cos(ax)} dx = -\frac{1}{a} \int \sec(ax) dx + \\ &\frac{1}{a} \int \cos(ax) dx \\ &= -\frac{1}{a^2} \log(\sec(ax) + \tan(ax)) + \frac{1}{a^2} \sin(ax) + C_1 \text{ -----(2)} \end{aligned}$$

$$B = \int \frac{xf}{w} dx + C_2, B = \int \frac{\tan(ax)\cos(ax)}{a} dx = \frac{1}{a} \int \sin(ax) dx = -\frac{1}{a^2} \cos(ax) + C_2 \text{ -----(3)}$$

Substitute (2) and (3) in (1),

$$y = \left(-\frac{1}{a^2} \log(\sec(ax) + \tan(ax)) + \frac{1}{a^2} \sin(ax) + C_1 \right) \cos(ax) \\ + \left(-\frac{1}{a^2} \cos(ax) + C_2 \right) \sin(ax)$$

$$y = C_1 \cos(ax) + C_2 \sin(ax) - \frac{1}{a^2} \cos a(x) \log(\sec(ax) + \tan(ax))$$

Example 3- Solve: $\frac{d^2y}{dx^2} + 4y = \sec(2x)$ by method of variation of parameters.

Solution: A.E is $m^2 + 4 = 0$

$$m^2 = \pm 2i$$

$$y = A \cos(2x) + B \sin(2x) \text{ -----(1)}$$

$$\text{Let } f = \cos(2x) \text{ and } g = \sin(2x), w = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} \cos(2x) & \sin(2x) \\ -2 \sin(x) & 2 \cos(x) \end{vmatrix} = 2,$$

$$X = \sec(2x)$$

$$A = - \int \frac{Xg}{w} dx + C_1, A = - \int \frac{\sec(2x) \sin(2x)}{2} dx = -\frac{1}{2} \int \tan(2x) dx$$

$$= \frac{1}{4} \log(\cos(2x)) + C_1 \text{ -----(2)}$$

$$B = \int \frac{Xf}{w} dx + C_2, B = \int \frac{\sec(2x) \cos(2x)}{2} dx = \frac{1}{2} \int dx = x + C_2 \text{ -----(3)}$$

Substitute (2) and (3) in (1),

$$y = \left(\frac{1}{4} \log(\cos(2x)) + C_1 \right) \cos(2x) + (x + C_2) \sin(2x)$$

$$y = C_1 \cos(2x) + C_2 \sin(2x) + \frac{1}{4} \cos(2x) \log(\cos(2x)) + x \sin(2x)$$

Example 4- Solve: $\frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = -3 \sin(2x)$ by method of variation of parameters.

Solution: A.E is $m^2 - 5m + 6 = 0$

$$(m - 2)(m - 3) = 0, m = 2, 3$$

$$y = A e^{2x} + B e^{3x} \text{ -----(1)}$$

$$\text{Let } f = e^{2x} \text{ and } g = e^{3x}, w = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} e^{2x} & e^{3x} \\ 2e^{2x} & 3e^{3x} \end{vmatrix} = e^{5x}, X = -3\sin(2x)$$

$$A = - \int \frac{Xg}{w} dx + C_1, A = - \int \frac{-3 \sin(2x)e^{3x}}{e^{5x}} dx = 3 \int e^{-2x} \sin(2x) dx$$

$$= \frac{3}{8} e^{-2x} (-2 \sin(2x) - 2 \cos(2x)) + C_1 \text{ -----(2)}$$

$$B = \int \frac{Xf}{w} dx + C_2, B = \int \frac{-3 \sin(2x)e^{2x}}{e^{5x}} dx = -3 \int e^{-3x} \sin(2x) dx$$

$$= \frac{-3}{13} e^{-2x} (-3 \sin(2x) - 2 \cos(2x)) + C_2 \text{ -----(3)}$$

Substitute (2) and (3) in (1),

$$y = \left(\frac{3}{8} e^{-2x} (-2 \sin(2x) - 2 \cos(2x)) + C_1 \right) e^{2x}$$

$$+ \left(\frac{-3}{13} e^{-2x} (-3 \sin(2x) - 2 \cos(2x)) + C_2 \right) e^{3x}$$

$$y = C_1 e^{2x} + C_2 e^{3x} - \frac{3}{52} \sin(2x) - \frac{15}{52} \cos(2x)$$

Example 5- Solve: $\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} - 3y = 4e^{2x}$ by method of variation of parameters.

Solution: A.E is $m^2 + 2m - 3 = 0$

$$(m + 3)(m - 1) = 0, m = -3, 1$$

$$y = A e^{-3x} + B e^{1x} \text{ -----(1)}$$

$$\text{Let } f = e^{-3x} \text{ and } g = e^x, w = \begin{vmatrix} f & g \\ f' & g' \end{vmatrix} = \begin{vmatrix} e^{-3x} & e^x \\ -3e^{-3x} & e^x \end{vmatrix} = 4e^{-2x}, X = 4e^{2x}$$

$$A = - \int \frac{Xg}{w} dx + C_1, A = - \int \frac{4e^{2x}e^x}{4e^{-2x}} dx = - \int e^{5x} dx$$

$$= -\frac{1}{5} e^{5x} + C_1 \text{ -----(2)}$$

$$B = \int \frac{xf}{w} dx + C_2, B = \int \frac{4e^{2x}e^{-3x}}{4e^{-2x}} dx = \int e^{1x} dx$$

$$= e^x + C_2 \text{-----}(3)$$

Substitute (2) and (3) in (1),

$$y = \left(-\frac{1}{5}e^{5x} + C_1\right)e^{-3x} + (e^x + C_2)e^x$$

$$y = C_1e^{-3x} + C_2e^x + \frac{4}{5}e^{2x}$$

.....

Solution of simultaneous first order differential equations

The system of simultaneous linear differential equations which contain a single independent variable and two or more dependent variables. In general, the number of equations will be equal to the number of dependent variables i.e. if there are n dependent variables there will be n equations.

For example:

The following two-equation system where x and y are dependent variables and t is the independent variable:

$$2\frac{dx}{dt} + \frac{dy}{dt} - 3x + y = 5e^{-2t};$$

$$3\frac{dx}{dt} - 2x + 4y = \sin(t)$$

or, equivalently,

$$2(D - 3)x + (D + 1)y = 5e^{-2t}$$

$$(3D - 2)x + 4y = \sin(t)$$

where $D = d/dt$.

Example 1: Solve the simultaneous ordinary differential equations:

$$(D + 4)x + 3y = t, \quad 2x + (D + 5)y = e^{2t}$$

Solution:

$$\text{Given } (D + 4)x + 3y = t \rightarrow (1)$$

$$2x + (D + 5)y = e^{2t} \rightarrow (2)$$

$$2 \times (1) - (D + 4) \times (2)$$

$$6y - (D + 4)(D + 5)y = 2t - (D + 4)e^{2t}$$

$$[6 - D^2 - 9D - 20]y = 2t - 2e^{2t} - 4e^{2t}$$

$$(D^2 + 9D + 14)y = 6e^{2t} - 2t$$

$$\text{The A.E. is } m^2 + 9m + 14 = 0$$

$$(m + 7)(m + 2) = 0$$

$$m = -2, -7$$

$$\text{C.F.: } Ae^{-2t} + Be^{-7t}$$

$$\begin{aligned}
 \text{P.I.} &= \frac{6}{(D^2 + 9D + 14)} e^{2t} - \frac{2}{(D^2 + 9D + 14)} t \\
 &= \frac{6e^{2t}}{4 + 18 + 14} - \frac{2}{14} \frac{1}{1 + \frac{9D}{14} + \frac{D^2}{14}} (t) \\
 &= \frac{6e^{2t}}{36} - \frac{1}{7} \left(1 + \frac{9D}{14} + \frac{D^2}{14} \right)^{-1} (t) \\
 &= \frac{e^{2t}}{6} - \frac{1}{7} \left(1 - \frac{9D}{14} \right) (t) = \frac{e^{2t}}{6} - \frac{1}{7} \left(t - \frac{9}{14} \right)
 \end{aligned}$$

$$\text{G.S. is } y = Ae^{-2t} + Be^{-7t} + \frac{e^{2t}}{6} - \frac{t}{7} + \frac{9}{98}$$

To Calculate x

$$Dy = -2Ae^{-2t} - 7Be^{-7t} + \frac{2e^{2t}}{6} - \frac{1}{7}$$

$$5y = 5Ae^{-2t} + 5Be^{-7t} + \frac{5e^{2t}}{6} - \frac{5t}{7} + \frac{45}{98}$$

$$(D + 5)y = 3Ae^{-2t} - 2Be^{-7t} + \frac{7e^{2t}}{6} - \frac{5t}{7} - \frac{1}{7} + \frac{45}{98}$$

$$(2) \Rightarrow 2x = -(D + 5)y + e^{2t}$$

$$= -3Ae^{-2t} + 2Be^{-7t} - \frac{7e^{2t}}{6} + \frac{5t}{7} - \frac{31}{98} + e^{2t}$$

$$x = \frac{-3A}{2} e^{-2t} + Be^{-7t} - \frac{7}{72} e^{2t} + \frac{5t}{14} - \frac{31}{196}$$

The General solution is

$$x = \frac{-3A}{2} e^{-2t} + Be^{-7t} - \frac{e^{2t}}{12} + \frac{5t}{14} - \frac{31}{196}$$

$$y = Ae^{-2t} + Be^{-7t} + \frac{e^{2t}}{6} - \frac{t}{7} + \frac{9}{98}.$$

Example 2: Solve for x from the equations $x' - y = t$ and $x + y' = 1$.

Solution:

$$x' - y = t \rightarrow (1) \Rightarrow x' - y' = 1 \Rightarrow x' - 1 = y'$$

$$x + y' = 1 \rightarrow (2) \Rightarrow x + x' - 1 = 1$$

$$\text{Thus } x' + x = 2 \text{ (or)} (D^2 + 1)x = 2$$

$$m^2 + 1 = 0 \Rightarrow m = \pm i$$

$$\text{C.F. } A \cos t + B \sin t$$

$$\text{P.I.} = \frac{1}{(D^2 + 1)}(2) = (D^2 + 1)^{-1}(2) = 2$$

$$\text{G.S: } x = A \cos t + B \sin t + 2.$$

Practice Problems:

Solve:

$$1. \frac{dx}{dt} + y = e^t; x - \frac{dy}{dt} = t$$

$$2. \frac{dx}{dt} + 2x + 3y = 2e^{2t}; \frac{dy}{dt} + 3x + 2y = 0 \text{ (U.Q)}$$

$$3. \frac{dx}{dt} + 2y = \sin 2t; \frac{dy}{dt} - 2x = \cos 2t \text{ (U.Q)}$$

$$4. (D + 4)x + 3y = t; 2x + (D + 5)y = e^{2t} \text{ (U.Q)}$$

$$x = Be^{-7t} - \frac{3}{2}Ae^{-2t} - \frac{1}{12}e^{2t} + \frac{5t}{14} - \frac{31}{196}$$

$$5. \frac{dy}{dt} = x, \frac{dx}{dt} = y \text{ (U.Q)}$$

Answers:

$$x = A \cos t + B \sin t + \frac{e^t}{2} + t$$

$$y = \frac{e^t}{2} + A \sin t - B \cos t - 1$$

$$x = Ae^{-5t} + Be^t + \frac{8}{7}e^{2t}$$

$$y = Ae^{-5t} - Be^t + \frac{6}{7}e^{2t}$$

$$x = \frac{1}{2}(2B - 1) \cos 2t - A \sin 2t$$

$$y = A \cos 2t + B \sin 2t$$

$$y = Ae^{-2t} + Be^{-7t} + \frac{1}{6}e^{2t} + \frac{9}{98} - \frac{t}{7}$$

$$x = Ae^t - Be^{-t} \text{ \& } y = Ae^t - Be^{-t}$$

$$6. \frac{dx}{dt} - \frac{dy}{dt} + 2y = \cos 2t; \frac{dx}{dt} + \frac{dy}{dt} - 2x = \sin 2t \text{ (U.Q)} \quad x = e^t \left(A \cos t + B \sin t \right) - \frac{1}{2} \cos 2t$$

$$y = e^t \left(A \sin t - B \cos t \right) - \frac{1}{2} \sin 2t$$

$$7. \frac{dx}{dt} + 2x - 3y = t; \frac{dy}{dt} - 3x + 2y = e^{2t} \text{ (U.Q)}$$

$$x = Ae^{-5t} + Be^t + \frac{3}{7}e^{2t} - \frac{2}{5}t - \frac{13}{25}$$

$$y = Be^t - Ae^{-5t} + \frac{4}{7}e^{2t} - \frac{3}{5}t - \frac{12}{25}$$

$$8. \frac{dx}{dt} + y = \sin t; x + \frac{dy}{dt} = \cos t$$

$$x = 2 \cosh t \text{ \& } y = \sin t - 2 \sinh t$$

given that $x = 2$ and $y = 0$ at $t = 0$ (U.Q)

$$9. \frac{dx}{dt} - y = t \text{ \& } \frac{dy}{dt} + x = t^2$$

$$x = A \cos t + B \sin t + t^2 - 1$$

$$y = -A \sin 2t + B \cos t + 1$$

$$10. \frac{dx}{dt} + 5x - 2y = t \text{ \& } \frac{dy}{dt} + 2x + y = 0$$

$$x = \frac{-1}{27}(1+6t)e^{-3t} + \frac{1}{27}(1+3t)$$

so that $x = y = 0$ when $t = 0$

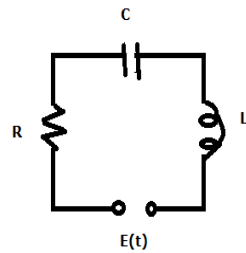
$$y = \frac{-1}{27}(2+3t)e^{-3t} + \frac{9}{27}(2-3t)$$

$$11. \frac{dx}{dt} + 2x + 3y = 2e^{2t}, \frac{dy}{dt} + 3x + 2y = 0$$

$$x = Ae^{-5t} + Be^t + \frac{8}{7}e^{2t}$$

$$y = Ae^{-5t} - Be^t - \frac{6}{7}e^{2t}$$

Applications: Electric Circuits



In a circuit containing inductance, resistance and capacitance, the charge q on the condenser is given by

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t) \rightarrow (1)$$

and the current i is given by

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = \frac{dE(t)}{dt} \rightarrow (2)$$

Where $E(t)$ is the applied electromotive force.

We shall discuss the solutions of the Equations (1) and (2) in this section.

It is interesting to note that there is striking similarity between Eq. (1) and the equations of oscillatory motions in mechanical systems are given below:

- (i) If $R=0$ and $E(t) = 0$, Eq. (1) is similar in form to the equation that represents free oscillations.
- (ii) If $E(t)=0$, Eq. (1) is similar in form to the equation that represents damped free oscillations.
- (iii) If $R=0$, $E(t) = E_0 \cos \omega t$, Eq. (1) is similar in form to the equation that represents forced oscillations without damping.
- (iv) If $E(t) = E_0 \cos \omega t$, Eq. (1) is similar in form to the equation that represents forced oscillations with damping.

Note: Eq. (2) is also to get the required result.

Example 1:

The charge q on a condenser satisfies the differential equation

$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = 0$ and $q(0) = Q$ and $q'(0) = 0$. Show that, if $4L > CR^2$, then $q = Q e^{-\alpha t} \left(\cos \omega t + \frac{\alpha}{\omega} \sin \omega t \right)$, where $\alpha = \frac{R}{2L}$ and $\omega^2 = \frac{4L - CR^2}{4L^2 C}$. Find the greatest negative charge at any time on the condenser.

Solution:

The given differential equation is

$$(LCD^2 + RCD + 1)q = 0 \rightarrow (1)$$

Where $D = d/dt$

The A.E of Eq. (1) is

$$LCm^2 + RCm + 1 = 0$$

$$\Rightarrow m = \frac{-RC \pm \sqrt{R^2 C^2 - 4LC}}{2LC}$$

$$\Rightarrow = -\frac{R}{2L} \pm i \sqrt{\frac{4LC - R^2 C^2}{4L^2 C^2}}, \text{ since } 4L > CR^2$$

$$\Rightarrow = -\frac{R}{2L} \pm i \sqrt{\frac{4L - R^2 C}{4L^2 C}} = -\alpha \pm i\omega$$

$a = LC$
 $b = RC$

Therefore, solution of Eq. (1) is

$$q = e^{-\alpha t} (A \cos \omega t + B \sin \omega t) \rightarrow (2)$$

when $t=0$, $q=Q$ in (2),

$$A=Q$$

Diff (2) w.r.t 't',

$$\frac{dq}{dt} = -\alpha e^{-\alpha t} (Q \cos \omega t + B \sin \omega t) + e^{-\alpha t} (-\omega Q \sin \omega t + \omega B \cos \omega t) \rightarrow (3)$$

When $t=0$, $\frac{dq}{dt} = 0$ in (3)

$$-\alpha Q + \omega B = 0$$

$$B = \frac{\alpha}{\omega} Q$$

Using the values of A and B in (2), we have

$$q = Q e^{-\alpha t} \left(\cos \omega t + \frac{\alpha}{\omega} \sin \omega t \right) \rightarrow (4)$$

From (3),

$$\frac{dq}{dt} = 0 \text{ if } -\alpha e^{-\alpha t} (Q \cos \omega t + \frac{\alpha}{\omega} Q \sin \omega t) + e^{-\alpha t} (-\omega Q \sin \omega t + \alpha Q \cos \omega t) = 0$$

$$\text{if } \left[\frac{-Q}{\omega} e^{-\alpha t} (\alpha^2 + \omega^2) \sin \omega t \right] = 0 \rightarrow (5)$$

if $\sin \omega t = 0$ since $e^{-\alpha t} = 0$ only when $t = \infty$

if $t=0$ and $\frac{\pi}{\omega}$

$$\frac{d^2q}{dt^2} = -\frac{Q}{\omega} (\alpha^2 + \omega^2) [-\alpha e^{-\alpha t} \sin \omega t + \omega e^{-\alpha t} \cos \omega t]$$

From (5),

$$\left(\frac{d^2q}{dt^2} \right)_{t=0} = -Q(\alpha^2 + \omega^2) < 0$$

$$\left(\frac{d^2q}{dt^2} \right)_{t=\frac{\pi}{\omega}} = Q(\alpha^2 + \omega^2)e^{-\frac{\alpha\pi}{\omega}} > 0$$

Therefore, the negative charge $(-q)$ is maximum at $t = \frac{\pi}{\omega}$

From (4), maximum value of $(-q) = Qe^{-\frac{\alpha\pi}{\omega}}$.

Example 2:

An uncharged condenser of capacity C is charged by applying an e.m.f E in $\left(\frac{t}{\sqrt{LC}}\right)$ through leads of self-inductance L and negligible resistance. Prove that, at time t , the charge q on one of the plates is

$$\frac{1}{2} EC \left[\sin \frac{t}{\sqrt{LC}} - \frac{t}{\sqrt{LC}} \cos \frac{t}{\sqrt{LC}} \right].$$

Solution:

Since the circuit contains only inductance and capacitance, the differential equation of the charge q on the condenser is

$$L \frac{d^2q}{dt^2} + \frac{1}{C} q = E(t), \text{ where } E(t) \text{ is the applied e.m.f.}$$

$$= E \sin \left(\frac{t}{\sqrt{LC}} \right)$$

$$\frac{d^2q}{dt^2} + \frac{1}{LC} q = \frac{E}{L} \sin \left(\frac{t}{\sqrt{LC}} \right) \rightarrow (1)$$

Putting $\frac{1}{LC} = \omega^2$, Eq. (1) can be written as

$$(D^2 + \omega^2)q = \frac{E}{L} \sin \omega t$$

The A.E of Eq. (1) is

$$m^2 + \omega^2 = 0$$

$$m = \pm i\omega$$

$$C.F = A \cos \omega t + B \sin \omega t$$

$$\begin{aligned} P.I &= \frac{1}{D^2 + \omega^2} \frac{E}{L} \sin \omega t \\ &= \frac{E}{L} \left(-\frac{t}{2\omega} \cos \omega t \right) \end{aligned}$$

Therefore, solution of Eq. (1) is

$$q = A \cos \omega t + B \sin \omega t - \frac{E t}{2L\omega} \cos \omega t \rightarrow (2)$$

Initially, when $t=0$, $q=0 \rightarrow (3)$

Using (3) in (2), we get $A=0$

Diff (2) w.r.t t , we have

$$\frac{dq}{dt} = -\omega A \sin \omega t + B \omega \cos \omega t - \frac{E}{2L\omega} \cos \omega t + \frac{E t}{2L} \sin \omega t \rightarrow (4)$$

Initially, when $t=0$, $i = \frac{dq}{dt} = 0 \rightarrow (5)$

Using (5) in (4), we have

$$B\omega - \frac{E}{2L\omega} = 0$$

$$B = \frac{E}{2L\omega^2} = \frac{EC}{2} \text{ since } \omega^2 = \frac{1}{LC}$$

Using the values of A and B in (2), we get

$$q = \frac{EC}{2} \sin \omega t - \frac{Et}{2L\omega} \cos \omega t$$

$$q = \frac{EC}{2} \left(\sin \omega t - \frac{t}{2LC\omega} \cos \omega t \right)$$

$$q = \frac{EC}{2} (\sin \omega t - \omega t \cos \omega t)$$

$$q = \frac{EC}{2} \left\{ \sin \frac{t}{\sqrt{LC}} - \frac{t}{\sqrt{LC}} \cos \frac{t}{\sqrt{LC}} \right\}$$

Example 3:

In an LCR circuit, the charge q on a plate of the condenser is given by $L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{q}{C} = E \sin \omega t$ Where $t = \frac{dq}{dt}$. The circuit is tuned to resonance so that $\omega^2 = \frac{1}{LC}$ if $R^2 < \frac{4L}{C}$ and $q=0=t$ when $t=0$. Show that

$$q = E/R\omega [-\cos \omega t + e^{(-Rt/2L)} (\cos pt + R/2Lp \sin pt)] \text{ and}$$

$$i = \frac{E}{R} \left[\sin \omega t - \frac{1}{p\sqrt{LC}} e^{-\frac{Rt}{2L}} \sin pt \right] \text{ where } p^2 = \frac{1}{LC} - \frac{R^2}{4L^2}.$$

Solution:

The given differential equation is

$$\left(LD^2 + RD + \frac{1}{C} \right) q = E \sin \omega t \quad \rightarrow (1)$$

where $D = \frac{d}{dt}$

The A.E of Eq. (1) is

$$L m^2 + Rm + \frac{1}{C} = 0$$

$$m = \frac{-R \pm \sqrt{R^2 - 4\frac{L}{C}}}{2L}$$

$$= -\frac{R}{2L} \pm i \sqrt{\frac{1}{LC} - \frac{R^2}{4L^2}} \quad \text{since } R^2 < \frac{4L}{C}$$

$$= -\frac{R}{2L} \pm ip \quad \text{since } p^2 = \frac{1}{LC} - \frac{R^2}{4L^2}$$

$$C.F = e^{-\frac{Rt}{2L}} \{A \cos pt + B \sin pt\}$$

$$P.I = \frac{1}{\left(LD^2 + RD + \frac{1}{C} \right)} E \sin \omega t$$

$$\begin{aligned}
&= \frac{1}{(-L\omega^2 + RD + \frac{1}{C})} E \sin \omega t \\
&= \frac{E}{R} \cdot \frac{1}{D} \sin \omega t \quad \left[L\omega^2 = \frac{1}{C} \right] \\
&= -\frac{E}{R\omega} \cos \omega t
\end{aligned}$$

The solution of Eq. (1) is

$$q = e^{\frac{-Rt}{2L}} \{A \cos pt + B \sin pt\} - \frac{E}{R\omega} \cos \omega t \rightarrow (2)$$

when $t=0$, $q=0$

$\rightarrow (3)$

Using (3) in (2), we have

$$A - \frac{E}{R\omega} = 0 \quad \text{or} \quad A = \frac{E}{R\omega}$$

Diff (2) w.r.t t , we get

$$i = \frac{dq}{dt} = -\frac{R}{2L} e^{\frac{-Rt}{2L}} \{A \cos pt + B \sin pt\} + p e^{\frac{-Rt}{2L}} \{-A \sin pt + B \cos pt\} + \frac{E}{R} \sin \omega t \rightarrow (4)$$

when $t=0$, $i=0$

$\rightarrow$

(5)

Using (5) in (4),

$$-\frac{R}{2L} A + p B = 0$$

$$B = \frac{R}{2Lp} \cdot \frac{E}{R\omega}$$

$$= \frac{E}{2\omega p L}$$

Using these values of A and B in (2), we get

$$\begin{aligned}
q &= e^{-\frac{Rt}{2L}} \left[\frac{E}{R\omega} \cos pt + \frac{E}{2\omega p L} \sin pt \right] - \frac{E}{R\omega} \cos \omega t \\
&= \frac{E}{R\omega} \left[-\cos \omega t + e^{-\frac{Rt}{2L}} \left(\cos pt + \frac{R}{2Lp} \sin pt \right) \right]
\end{aligned}$$

$\rightarrow (6)$

Diff (6) w.r.t 't', we get

$$\begin{aligned}
i = \frac{dq}{dt} &= \frac{E}{R} \sin \omega t - \frac{E}{2L\omega} e^{-\frac{Rt}{2L}} \left(\cos pt + \frac{R}{2Lp} \sin pt \right) \\
&\quad + \frac{E}{R\omega} e^{-\frac{Rt}{2L}} \left(-p \sin pt + \frac{R}{2L} \cos pt \right) \\
&= \frac{E}{R} \sin \omega t + \frac{E}{2L\omega} e^{-\frac{Rt}{2L}} (\cos pt - \cos \omega t) - \frac{E}{R} e^{-\frac{Rt}{2L}} \left(\frac{R}{4L^2 p} + \frac{p}{R} \right) \sin pt
\end{aligned}$$

$$= \frac{E}{\omega} \sin \omega t - \frac{E}{\omega} e^{-\frac{Rt}{2L} \frac{(R^2 + 4L^2 p^2)}{4L^2 p R}} \sin pt$$

$$= \frac{E}{R} \sin \omega t - \frac{E}{\omega} e^{-\frac{Rt}{2L}} \frac{4L}{C} \cdot \frac{1}{4L^2 p R} \sin pt$$

Since $p^2 = \frac{1}{LC} - \frac{R^2}{4L^2}$ and so $R^2 + 4L^2 p^2 = 4 \frac{L}{C}$

$$= \frac{E}{R} \sin \omega t - \frac{1}{LCpR\omega} e^{-\frac{Rt}{2L}} \sin pt$$

$$= \frac{E}{R} \left[\sin \omega t - \frac{\omega}{p} e^{-\frac{Rt}{2L}} \sin pt \right], \text{ since } \omega^2 = \frac{1}{LC}$$

$$= \frac{E}{R} \left[\sin \omega t - \frac{1}{p\sqrt{LC}} e^{-\frac{Rt}{2L}} \sin pt \right]$$

Example 4:

In an LRC-circuit, the impressed voltage is $400 \cos 250 t$ volts. Find the current in the circuit when $t=0.001$ second, if $L=1$ henry, $R=400$ ohms and $C = 0.16 \times 10^{-4}$ farad and the initial charge and current are zero.

Solution:

The differential equation satisfied by the charge q is

$$L \frac{d^2 q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E(t)$$

$$\frac{d^2 q}{dt^2} + 400 \frac{dq}{dt} + \frac{1}{0.16 \times 10^{-4}} q = 400 \cos 250t$$

$$(D^2 + 400D + 62,500)q = 400 \cos 250t$$

→ (1)

The A.E of Eq. (1) is

$$m^2 + 400m + 62,500 = 0$$

$$m = \frac{-400 \pm \sqrt{1,60,000 - 2,50,000}}{2}$$

$$= \frac{-400 \pm i300}{2}$$

$$= -200 \pm i150$$

$$C.F = e^{-200t}(A \cos 150 t + B \sin 150 t)$$

$$\begin{aligned} P.I &= \frac{1}{D^2 + 400D + 62,500} 400 \cos 250 t \\ &= \frac{1}{D} \cos 250 t \\ &= \frac{1}{250} \sin 250t \end{aligned}$$

Therefore, the solution of Eq. (1) is

$$(2) \quad q = e^{-200t}(A \cos 150t + B \sin 150t) + \frac{1}{250} \sin 250t \quad \rightarrow$$

$$\text{when } t=0, q=0 \quad \rightarrow (3)$$

Using (3) in (2), we get A=0

Using this value of A in (2) and diff w.r.t 't', we have

$$(4) \quad i = \frac{dq}{dt} = -200 e^{-200t} \cdot B \sin 150t + 150 B e^{-200t} \cos 150t + \cos 250t \quad \rightarrow$$

$$\text{when } t=0, i=0 \quad \rightarrow (5)$$

Using (5) in (4), we get

$$150B + 1 = 0$$

$$B = -\frac{1}{150}$$

Using the value of B in (4), we get

$$i = \frac{4}{3} e^{-200t} \sin 150t - e^{-200t} \cos 150t + \cos 250t$$

$$\begin{aligned} [i]_{t=0.001} &= \frac{4}{3} e^{-0.2} \sin (0.15) - e^{-0.2} \cos (0.15) + \cos (0.25) \\ &= 0.3225 \text{ ampere} \end{aligned}$$

Practice Problems:

1. In an L-C-R circuit, the charge q on a plate of a condenser is given by

$$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = E \sin pt.$$

The circuit is tuned to resonance so that $p^2 = 1/LC$. If initially the current I and the charge q be zero, show that, for small values of R/L , the current in the circuit at time t is given by $(Et/2L) \sin pt$.

2. A condenser of capacity C discharged through an inductance L and resistance R in series and the charge q at time t satisfies the equation

$L \frac{d^2q}{dt^2} + R \frac{dq}{dt} + \frac{1}{C} q = 0$. Given that $L=0.25$ henries, $R=250$ ohms, $C=2 \times 10^{-6}$ farads, and that when $t=0$, charge q is 0.002 coulombs and the current $dq/dt=0$, obtain the value of q in terms of t .
